ZentroMart Assist VoiceOps 2026

Capstone implementation report | Offline evidence and remaining live checks

Submission status
Runnable source prototype published. Offline checks passed. Live LLM and voice service runs remain unverified because local API credentials and a user recording were not available.

1. Project overview

ZentroMart Assist is a customer-support prototype for policy questions and order-status requests. It retrieves current sections from the supplied knowledge base, passes grounded context to an OpenAI-compatible chat-completions endpoint, and offers synthetic order lookup and escalation-ticket tools. The audio batch path uses Gnani Vachana for speech-to-text and text-to-speech. The repository contains source code and tests; the supplied synthetic dataset is intentionally kept out of the public repository.

2. Design and safeguards

Knowledge retrieval. Markdown documents are split by section identifier and ranked with a BM25-style lexical scorer. Superseded documents are excluded, and document-type precedence contributes to ranking. Retrieved source identifiers are attached to answer records.

Order privacy. The mock lookup tool asks for an order identifier and then validates a registered phone suffix or delivery pincode before returning private order facts. Ticket creation validates order status and summary content. These are educational synthetic-data controls, not production authentication.

Voice path. A directory batch runner accepts supported audio files, transcribes each, sends the transcript through the support agent, synthesizes the reply, and writes per-file text and JSON output. No personal recording is included in this submission.

3. Verification results

Check	Result	Evidence / limit
Dataset integrity	PASS	9 CSVs and 21 knowledge documents validated; 1,000 orders, 1,783 items, 244 logs, 52 questions, 23 conversations.
Unit tests	PASS: 3 tests	Current-section retrieval, verified order-data gating, and source-ID attachment. Offline tests use a mock LLM client.
Python syntax	PASS	Python modules compiled successfully.
Retrieval-only practice set	72.3% at top 8	34 of 47 cases with KB gold evidence retrieved at least one gold section in the top 8. This is retrieval coverage, not answer accuracy.
LLM answer evaluation	NOT RUN	OPENAI_API_KEY was not available locally. Baseline-vs-RAG answer scoring and conversation-suite evaluation require a local credential.
Gnani STT/TTS smoke test	NOT RUN	GNANI_API_KEY and a short recording of the user speaking were unavailable. No recording was fabricated or uploaded.

Check	Result	Evidence / limit
CI configuration	NOT CONFIGURED	No GitHub Actions workflow or other CI job is present in the repository; tests were run locally.

4. Interpretation

The offline evidence supports a working source prototype and verifies several core safeguards. The retrieval score indicates headroom remains: about one in four answerable gold-evidence cases did not return an expected section within the top eight. No claim is made about live answer quality, speech recognition quality, or spoken-output quality until the credentialed runs are completed.

5. Reproducibility and data handling

The README documents setup, local environment variables, chat, batch audio, and evaluation commands. Copy .env.example to .env and add credentials locally; .env, API keys, data folders, generated runtime outputs, and audio files are ignored by Git. The public repository contains no supplied dataset or personal recording.

Remaining steps

To complete the live evaluation, set your own LLM and Gnani credentials in a local .env file, run the three evaluation commands and the batch voice smoke test with a brief recording of your own voice, then append the resulting metrics and voice evidence. The capstone brief did not provide a submission portal URL, so portal submission is outside this repository handoff.

Repository: https://github.com/chaityy/zentromart-voiceops-2026